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Mulitple Views

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SET09123 – Interactive Data Visualisation

Lecture Summary

- Superimposition
- Juxtaposition
  - Small Multiples
- Dashboard Design

View Multiplicity

Motivation

Why would you want multiple views?

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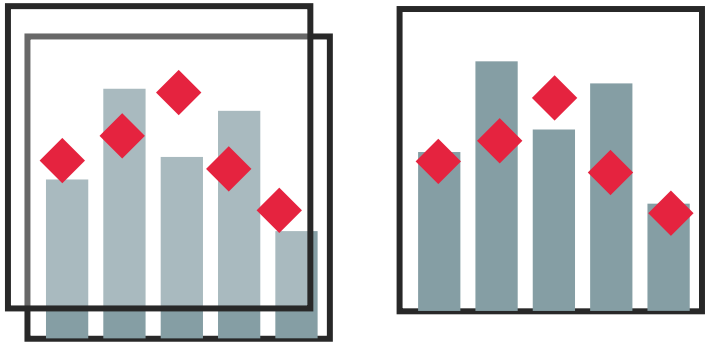
Why would you want multiple views?

- Again, to deal with *complex and large* datasets
  - We can split different facets of the dataset into different views
- Similarly, different views can support *different user tasks*

Arranging Multiple Views

- Superimposition
  - Layers arranged on top of each other



Arranging Multiple Views

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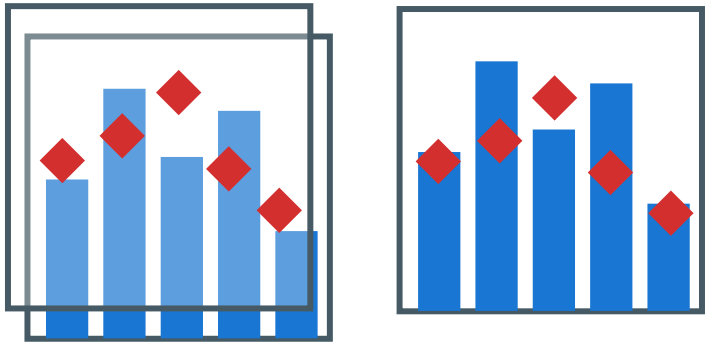
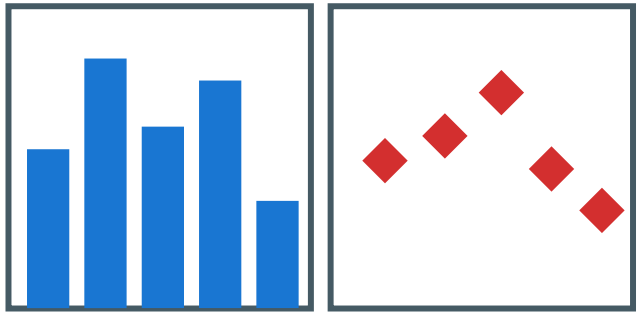


- Juxtaposition
  - Views arranged side-by-side



Superimposition

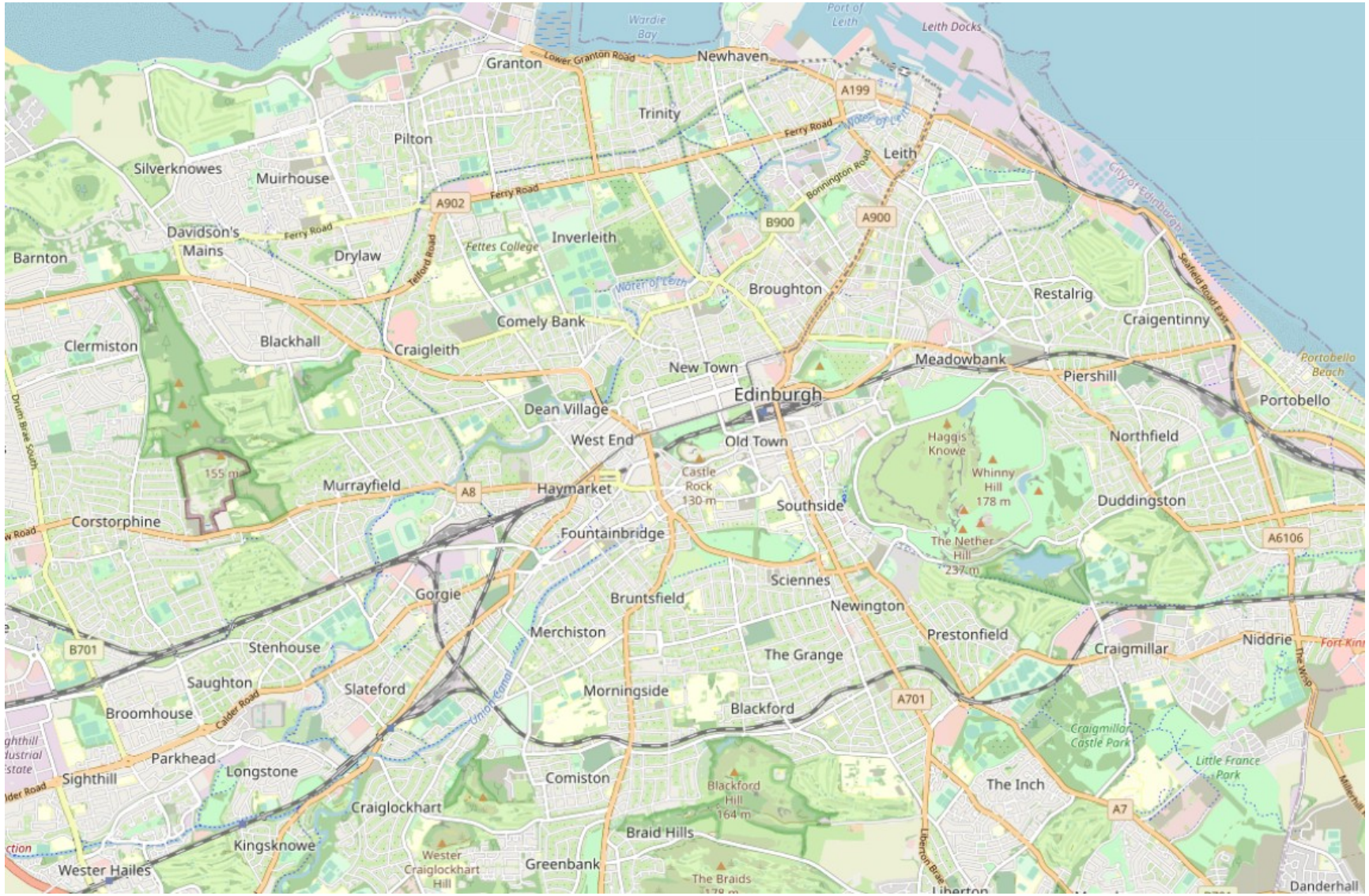
Superimposition

- Consists of superimposing, or stacking, multiple layers of data
- Each layer is a **group of related marks**
- All layers spread across the **same visual range**
  - Although they can have different domain
  - Hence several scales (and axes) can be used
- Typically involves background and foreground layers

Considerations

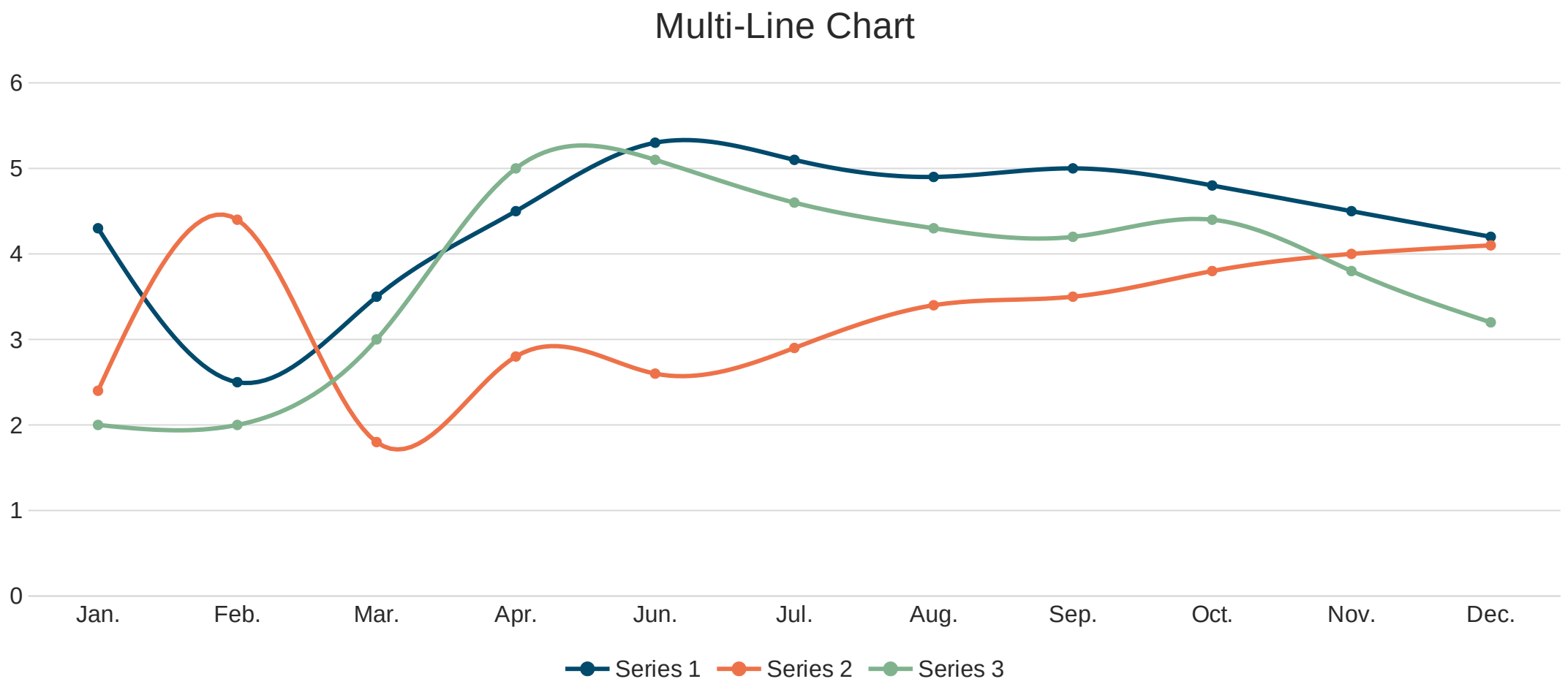
- For readability, layers have to be visually distinguishable
  - Especially for static / non-interactive charts
- Make sure they don’t use the similar or overlapping visual channels
- Unless the visual channel is meant to differentiate layers, e.g. colour hue

Example



Source: OpenStreetMap

Example


Multi-Line Chart
6
5
4
3
2
1
0
Jan.
Feb.
Mar.
Apr.
Jun.
Jul.
Aug.
Sep.
Oct.
Nov.
Dec.
Series 1
Series 2
Series 3

Usage

- Superimposition easily supports tasks like comparison
  - E.g., with multi-line charts
- It’s also a mechanism for embedding *focus and context* together
  - E.g., the superimposition of a minimap (context) on top of a zoomed-in map (focus)

Juxtaposition

Juxtaposition

- Superimposition can quickly reach visual clutter
- Juxtaposition puts the views side-by-side
- Three considerations:
  - Sharing data subsets
  - Sharing visual encoding
  - Sharing interaction

Juxtaposition Types

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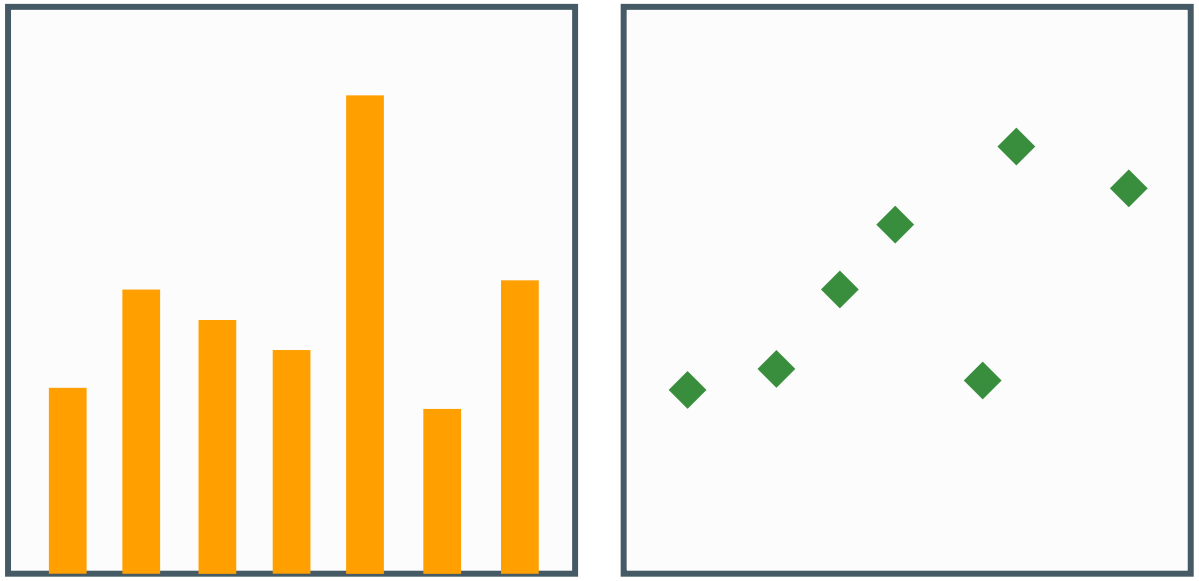
		Data Shared		
		All	Subset	None
Encoding Shared	Yes	*Redundant*	Overview/Detail	Small Multiples
	No	Multiform	Multiform Overview/Detail	*No linkage*

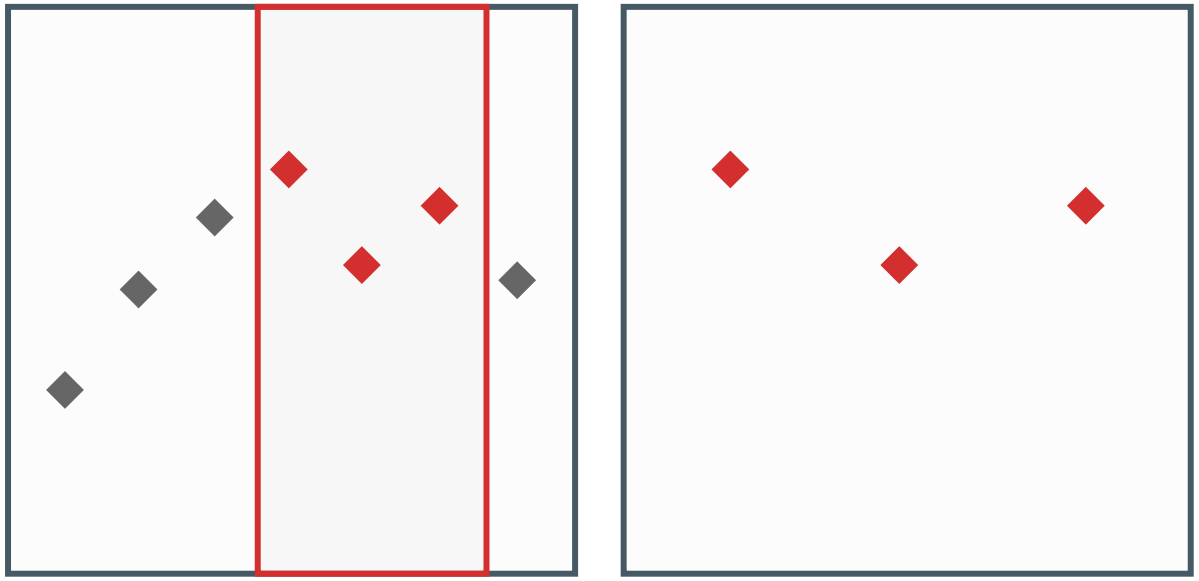
Multiform

- Multiform exploits the advantages of using different encoding channels to show the different facet or pattern of the data



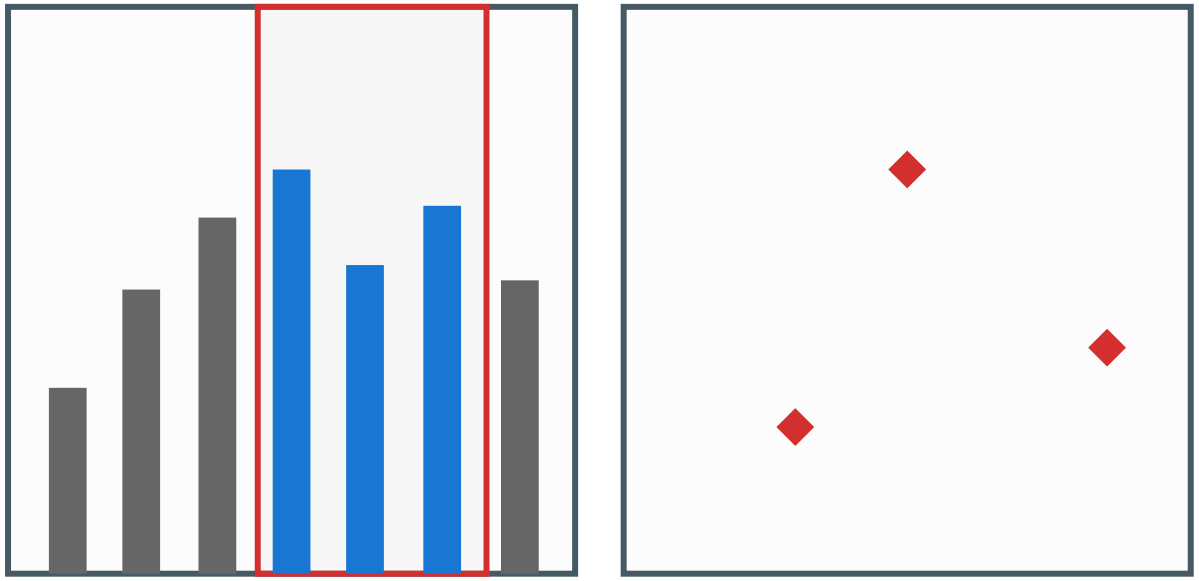
Overview/Details

- Similar to the *focus & context* paradigm sometimes seen with superimposition
- One view shows the full extent of the data, the other view concentrates on a subset



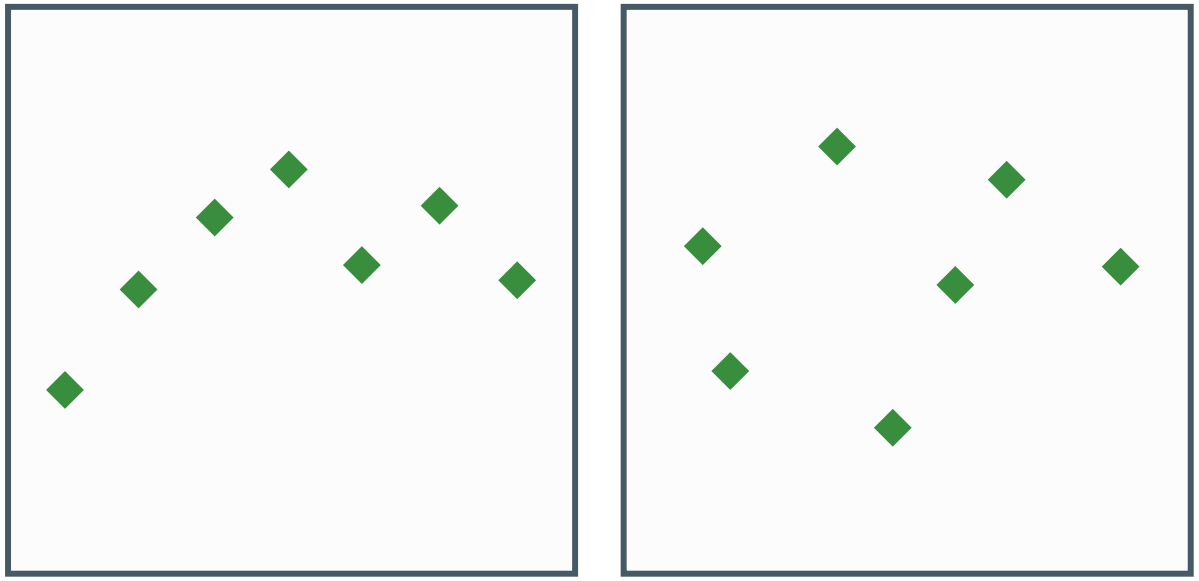
Multiform Overview/Details

- The two paradigms, multiform and overview/details, can be used in conjunction
- In practice, one view is used to filter the data of another



Small Multiples

- Each view shows a different facet of the data, using the same visual encoding
- Small multiples are typically aligned in lists or grids to facilitate comparison



Small Multiples

- Typically, Small Multiples are a good alternative to animated views
- All facets of the data are visible and accessible to the user to glance between them with minimal interaction costs and memory load
- However, Small Multiple rapidly increase the screen real-estate required

Coordination

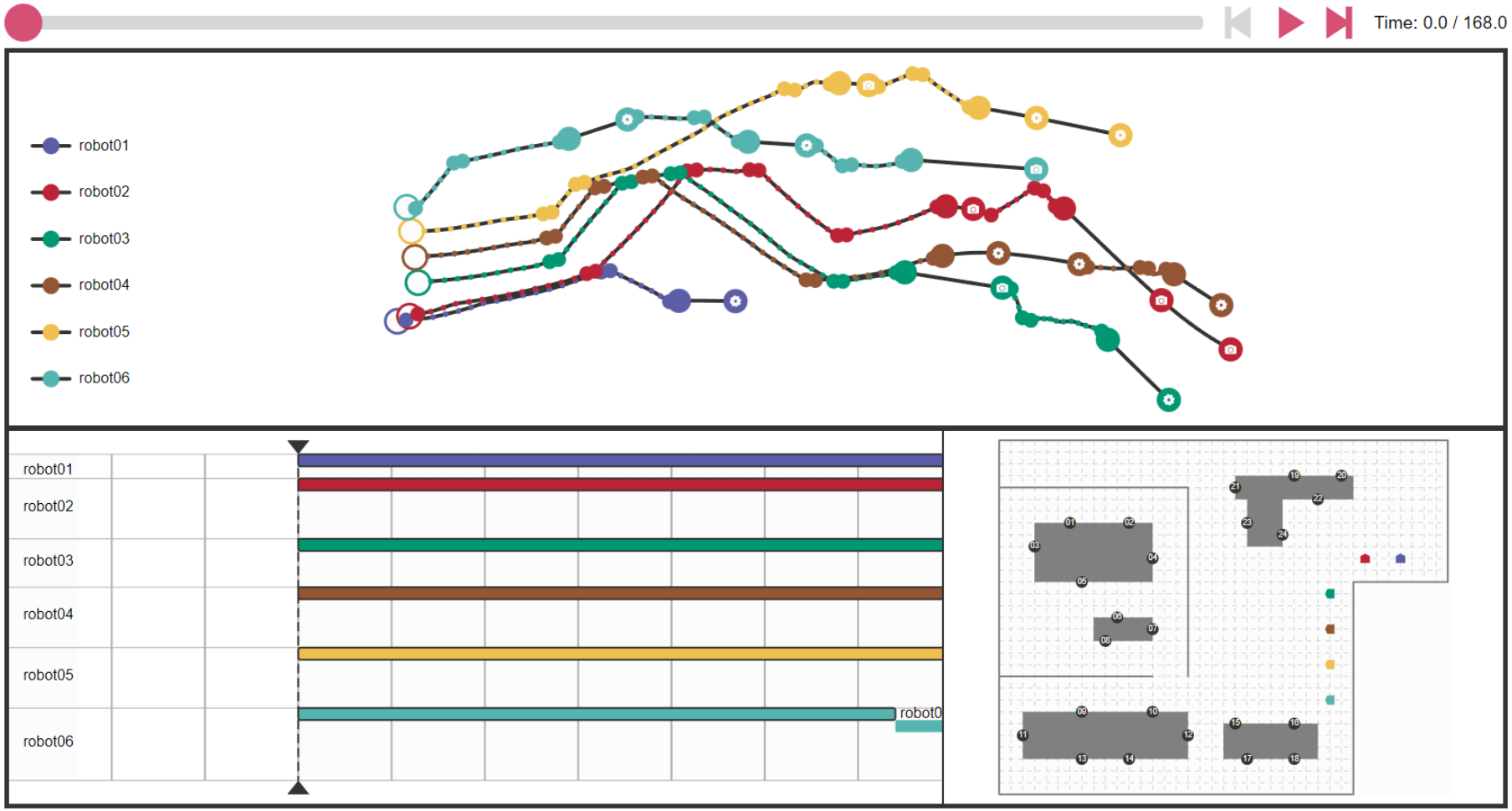
- Juxtaposed views are often **coordinated** by sharing their interaction(s)

- *Linked selection* – selection in one view is reflected in other views
  - E.g., by selecting marks or reducing data

- *Linked highlight* – marks representing the same item(s) are highlighted across all views
  - Bidirectional linked highlight is more difficult to implement but preferable

- *Linked navigation* – synchronising navigation between views
  - Map / minimap panning and zooming coordination
  - Shared scrolling across timeseries visualisations

Example


Time: 0.0 / 168.0
robot01
robot02
robot03
robot04
robot05
robot06
robot01
robot02
robot03
robot04
robot05
robot06
robot0

Limitation

- Multiplying views can easily lead to a *cognitive overload* for the user
  - There's a lot to pay attention to and memorise
- Inappropriate/non-existing coordination can simply confuse and frustrate them
- Carefully considering the design of your *dashboard* is essential

Dashboard Design

What is a Dashboard?

- Typically a *single page* interface
- Provides an *at-a-glance view* of the data across *multiple views*
- Enables *rapid interaction* to respond to user queries

Dashboard Purposes

- Operational: helps user day-to-day tasks
  - Often includes many options for immediate actions
- Strategic: present the data for decision-making
  - Would support interactions to explore the data
- Communication: simply present the data
  - Includes little or no interaction

Sarikaya, Correll, M., Bartram, L., Tory, M., & Fisher, D. (2019). What Do We Talk About When We Talk About Dashboards? *IEEE Transactions on Visualization and Computer Graphics*, 25(1), 682–692. https://doi.org/10.1109/TVCG.2018.2864903

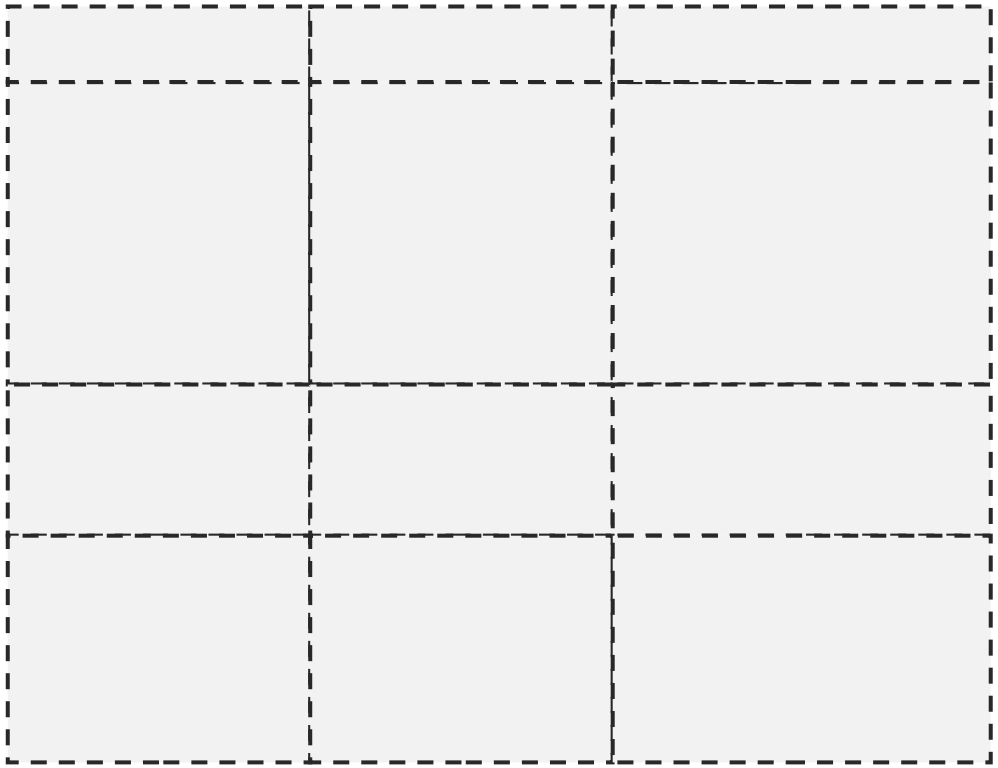
Dashboard Audience

- Who gets to see the dashboard?
  - General Public, Social Circles, Organisations/Business, Individuals
- Are they visualisation-literate?
- Are they domain expert?

Sarikaya, Correll, M., Bartram, L., Tory, M., & Fisher, D. (2019). What Do We Talk About When We Talk About Dashboards? *IEEE Transactions on Visualization and Computer Graphics*, 25(1), 682–692. https://doi.org/10.1109/TVCG.2018.2864903

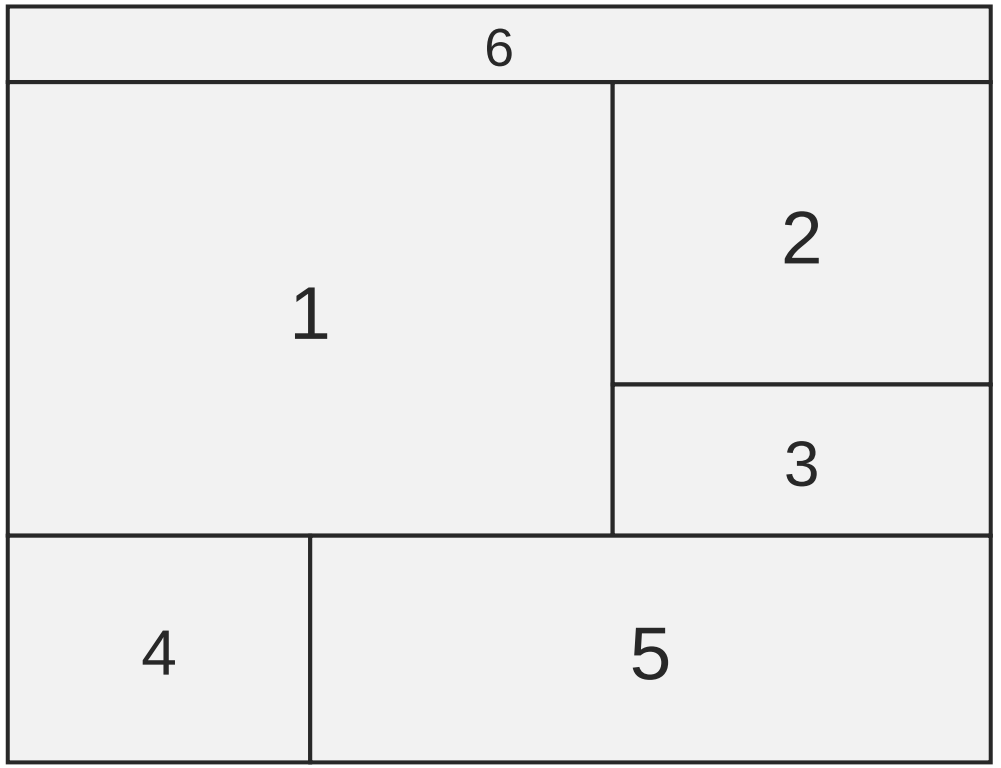
Dashboard Design

- Users are better at parsing regular layouts
  - Make sure to keep the views aligned
  - Using an underlying grid is usually good practice



Dashboard Design

- There is a hierarchy in which users prioritise content
  - In western cultures (and most computer UI), left to right and top to bottom
  - Size also has an influence



Dashboard Design

- Gestalt law of proximity: items which are close are perceptually grouped together
- Include white-space to infer which views work closely together

Other Considerations

- Interaction needs to be *reactive*
  - Implement immediate visual feedback as much as possible

- Dashboards present an *overview* of the data
  - And you are limited in screen real-estate
  - Consider the most important *user tasks* to support
  - You can always create other, more specialised, pages

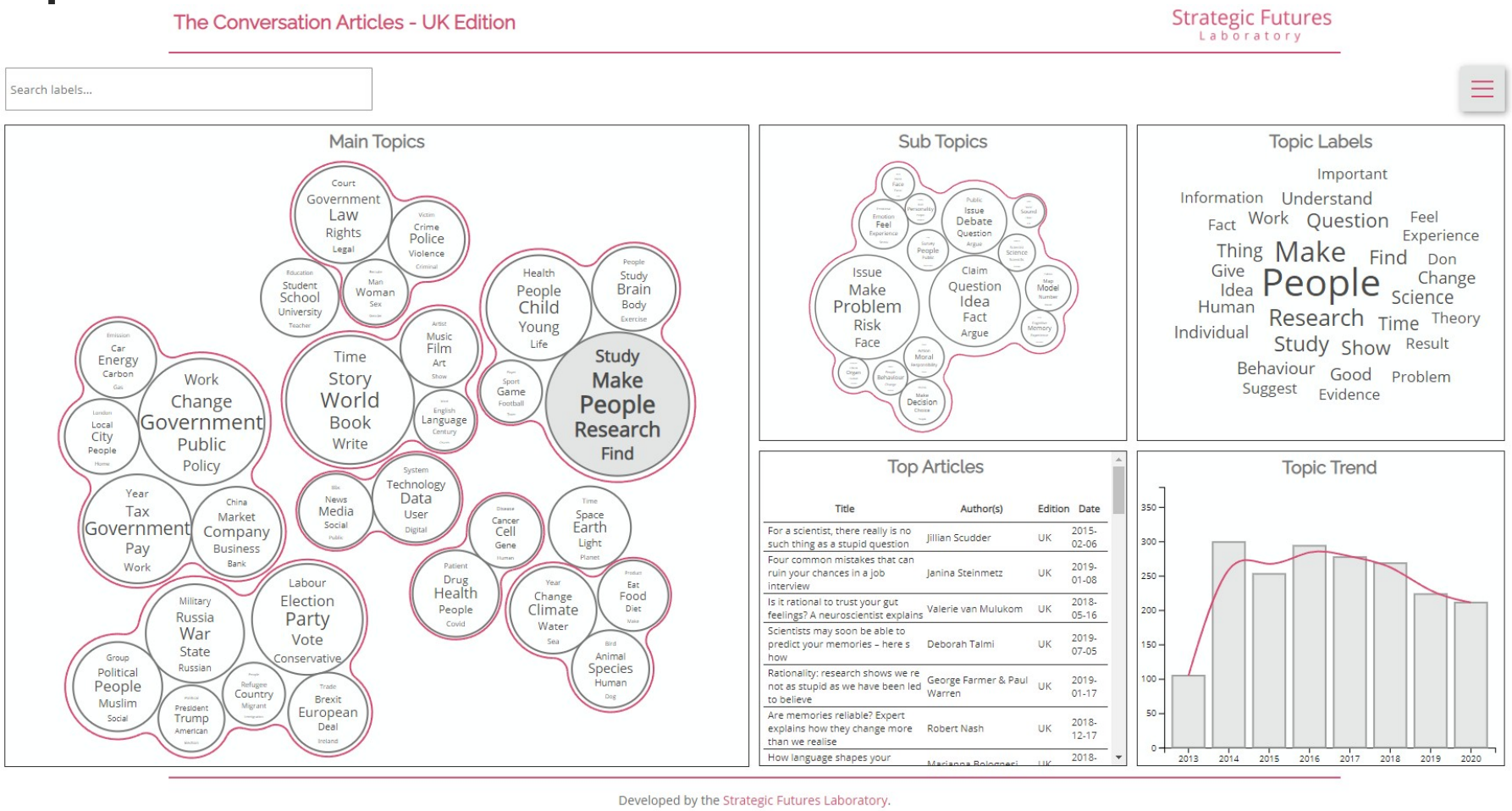
- Multiple views are cognitively demanding
  - Limit the efforts you ask of your users, i.e. two different representations for the same type of data

Final Advice

- Much like data analysis, design is an iterative process

- Create your dashboard and have tested by users to check:
  - How well/bad do they manage to perform their task?
  - Have you overlooked an important task?
  - Have you put too much emphasis on a minor task?
  - Are there common, less-important, information you can represent?
  - Are there interactions/visualisations that could further aid users in their task?

Example



https://strategicfutures.org/TopicMaps/TheConversation/ukEdition.html

Conclusion

Key Points

- Another way to deal with complexity in Data Visualisation is to rely on multiple views

- **Superimposition** shows views as stacked layers
  - Which can be useful for spatial comparison

- **Juxtaposition** arranges views side-by-side
  - Small Multiples are a well-known paradigm to help the analysis of multidimensional datasets

- Careful placement and interaction coordination is key to effective **dashboards**